

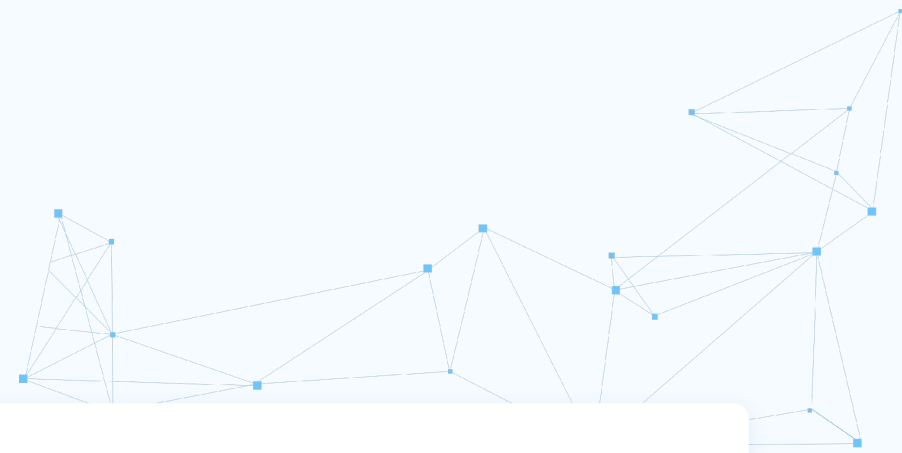
AI-Assisted R Package Development

From an idea to something you would let other people depend on

Validated AI for Pharma Summit

Vedha Viyash · Appsilon · 23rd September 2026

Where we're going



01

How R package development evolved

Slides and polls. Every step of building a package, before AI and now, and the parts that did not move.

the workflow

what to delegate

what not to

02

Idea to package, *live* (kinda)

An empty repo, an idea, and an agent. Let's go through the process together.

architecture

tests first

the harness

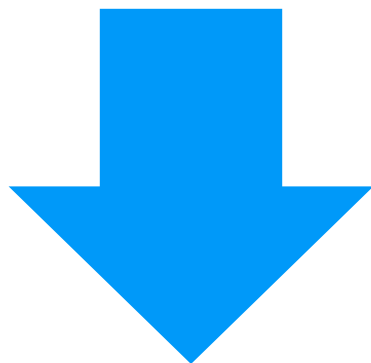
Quick poll 



<https://vedhav.github.io/live-survey>

Two costs, and only one of them moved

COST OF WRITING A PACKAGE



Collapsed

Scaffolding, docs, tests, boilerplate.

COST OF BEING WRONG



Exactly what it always was

A broken contract, a wrong number, a lost user.

Everything in this session follows from that gap.

What you used to have to learn

DESCRIPTION

NAMESPACE

roxygen2

testthat

R CMD check

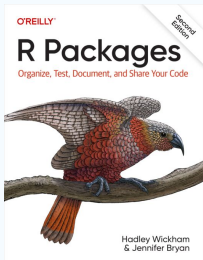
CRAN policy

pkgdown

NEWS.md

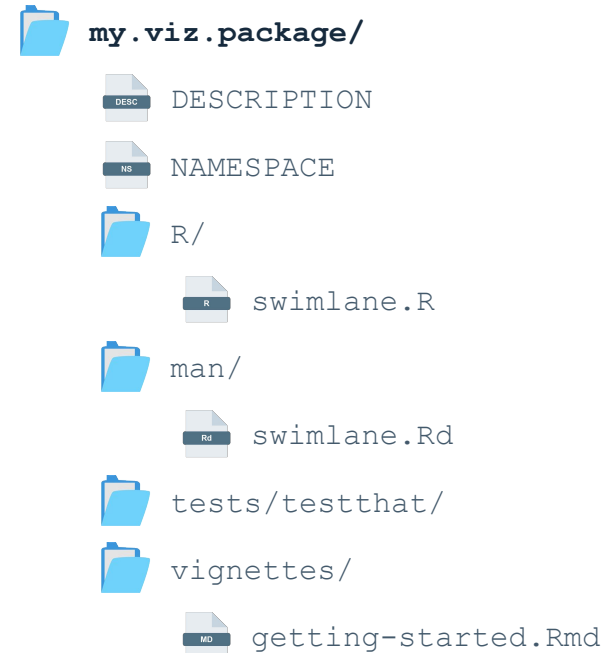
vignettes

S3 / S4 / R6



Weeks of it, before you shipped anything anybody wanted.

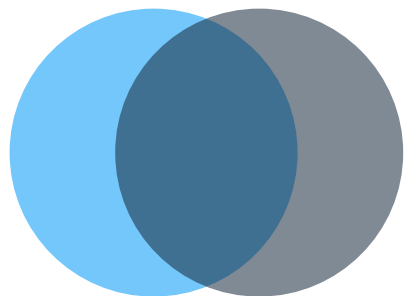
AND THE SHAPE IT ALL HAS TO TAKE



What is the most important thing **code** provides us ?

Mental alignment

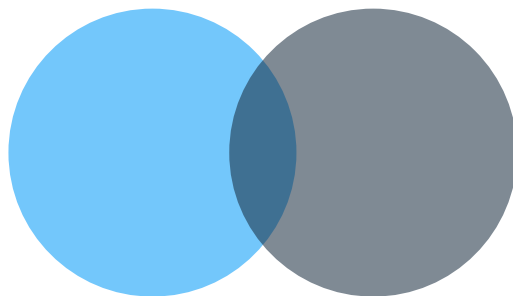
Writing code was never only about *producing code*. It was **how the repo and your head stayed in sync**.



Aligned

You write it

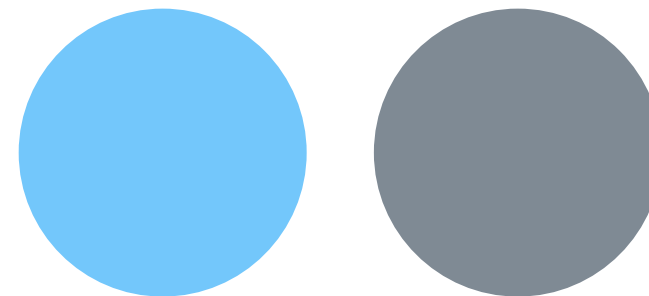
You cannot type it without understanding it.



Drifting

You delegate and review

Most of it lands. Some of it you took on trust.



Gone

You delegate and skim

It passes. You could not defend a line of it.

 What the code does

 What you think it does

Two functions. Same happy path.

```
f <- function(x, y) {  
  paste(x, y, sep = "T")  
}
```

```
derive_iso_datetime <- function(date, time = NULL) {  
  if (is.null(time) || is.na(time) || time == "") {  
    return(date)  
  }  
  
  paste0(date, "T", time)  
}
```

INPUT

VERSION A

VERSION B

time = "08:30"

"2024-03-01T08:30"

"2024-03-01T08:30"

identical

time = NA

"2024-03-01TNA"

"2024-03-01"

different

time = ""

"2024-03-01T"

"2024-03-01"

different

The happy-path hides the difference.

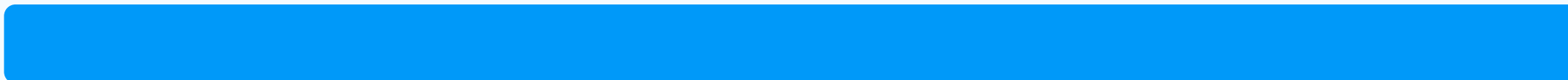
Both look right until the time is missing, and a missing time is normal. Robust tests are how you find that.

What that means now?

You can now ship code you don't understand, faster than ever before...



What you shipped

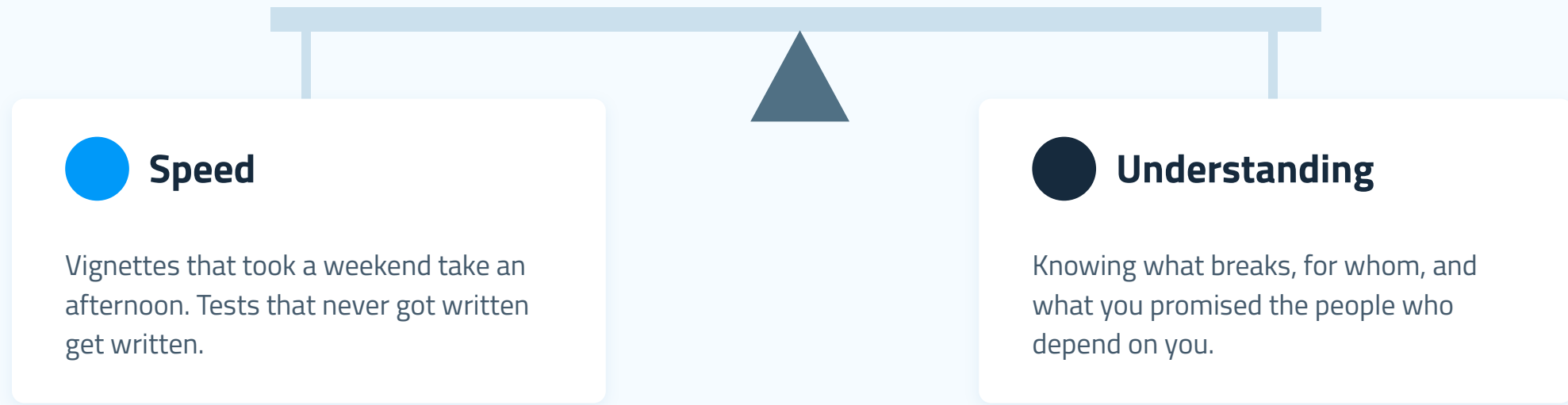


What you could defend in a review



That gap is the risk, and nothing in the toolchain closes it for you.

This is not an anti-AI talk



Both are real. The work is finding where the balance sits for the task in front of you.

Delegate what you can judge ❌🤔✅

Can you judge the output?



Yes

Hand it over

Give the agent the task, then check the result against something you can actually verify.



No

Understand it first

Research, ask, read, plan. Only then delegate the implementation. Skipping this is how you lose alignment.

This works wherever you are on the curve. It just changes your opening move.

The same task, two prompts

PROMPT A

```
Write a function to summarise  
adverse events by treatment arm.
```

Nine decisions left to the agent

Every one of these gets decided either way:

input datasets

subjects or events

denominator

population flag

missing arm

output type

sort order

rounding

term level

The only question is who made them.

PROMPT B

```
From ADAE and ADSL, count distinct  
subjects with at least one event, by arm  
and preferred term. Denominator is the  
safety population in ADSL. Return a  
tibble sorted by descending count,  
percentages to 1 dp. A missing arm is  
an error, not a silent drop.
```

Nine decisions made by me

More freedom, more average

The more choices you leave open, the closer the answer lands to the most common one.

Fine when your case is typical. Quietly wrong when it is not, and it never announces which happened.

WHAT TO TAKE AWAY

Unstated constraints are unconsidered constraints

If you never said the safety population was the denominator, it was never weighed. Not refused, not flagged. Just never considered.

A COMMON CAVEAT

Careful with “regression to the mean”

It is a useful shorthand, but the precise claim is about the modal reading of an ambiguous request, not about the model being mediocre.

The whole job

Mechanics delegate. Decisions don't.

Six places we're going deep

Agent

Agent-led

Shared

You-led

You

Architecture

class or function?

API design & exports

the contract

Testing

behaviour vs mirror

Documentation

the biggest win

Release & deprecation

keeping promises

Regulated / GxP

validation evidence

Same three questions each time: how it was done, what an agent does now, and what didn't move.

Architecture

STEP	WITH AI TODAY	LEVEL
Scaffolding	usethis, instantly	Agent
Dependency choice	Lists options and trade-offs fast	Shared
Package shape	Implements whatever you name	You
Class vs function	Defaults to plain functions	You

WHAT STAYS HUMAN

The shape of the thing

An agent will not invent a constraint you never mentioned. Multi-engine was my decision, and it changed everything downstream.

API design & exports

STEP	WITH AI TODAY	LEVEL
Input validation	Thorough, fast, good messages	Agent-led
Naming & signatures	Proposes consistent families	You-led
Return contract	Invents something plausible	You-led
Export vs internal	Over-exports by default	You

WHAT STAYS HUMAN

Every export is a contract

Users build on it. Break it and you don't lose a function, you lose the trust that made them depend on you.

Testing

STEP	WITH AI TODAY	LEVEL
Unit tests	Writes many, very fast	Shared
Fixtures	Generates plausible data	Shared
Coverage	Raises the number quickly	Shared
Visual regression	Builds the harness, can't judge	You

WHAT STAYS HUMAN

Behaviour, not mirror

A test written from the implementation passes by construction. Coverage is a proxy, and an agent will happily game it.

Documentation

STEP	WITH AI TODAY	LEVEL
Roxygen blocks	Drafts complete blocks	Agent-led
@examples	Runnable, and actually written	Agent-led
Vignettes	Structure and prose drafted	Shared
NEWS.md	Drafted straight from the diff	Shared

WHAT STAYS HUMAN

The biggest honest win

Vignettes were the bottleneck and now they aren't. What stays yours: whether it says the true thing, and which example earns the space.

Release & deprecation

STEP	WITH AI TODAY	LEVEL
Deprecation mechanics	Badge, warning, NEWS, timeline	Agent
Version bump	Applies the rule you state	Shared
Breaking-change call	Doesn't know your users	You
Backward compatibility	Breaks it silently unless told	You

WHAT STAYS HUMAN

The promise you made

Mechanics are perfectly delegable.
Whether to deprecate, and how
much notice you owe people, is not.

Regulated / GxP

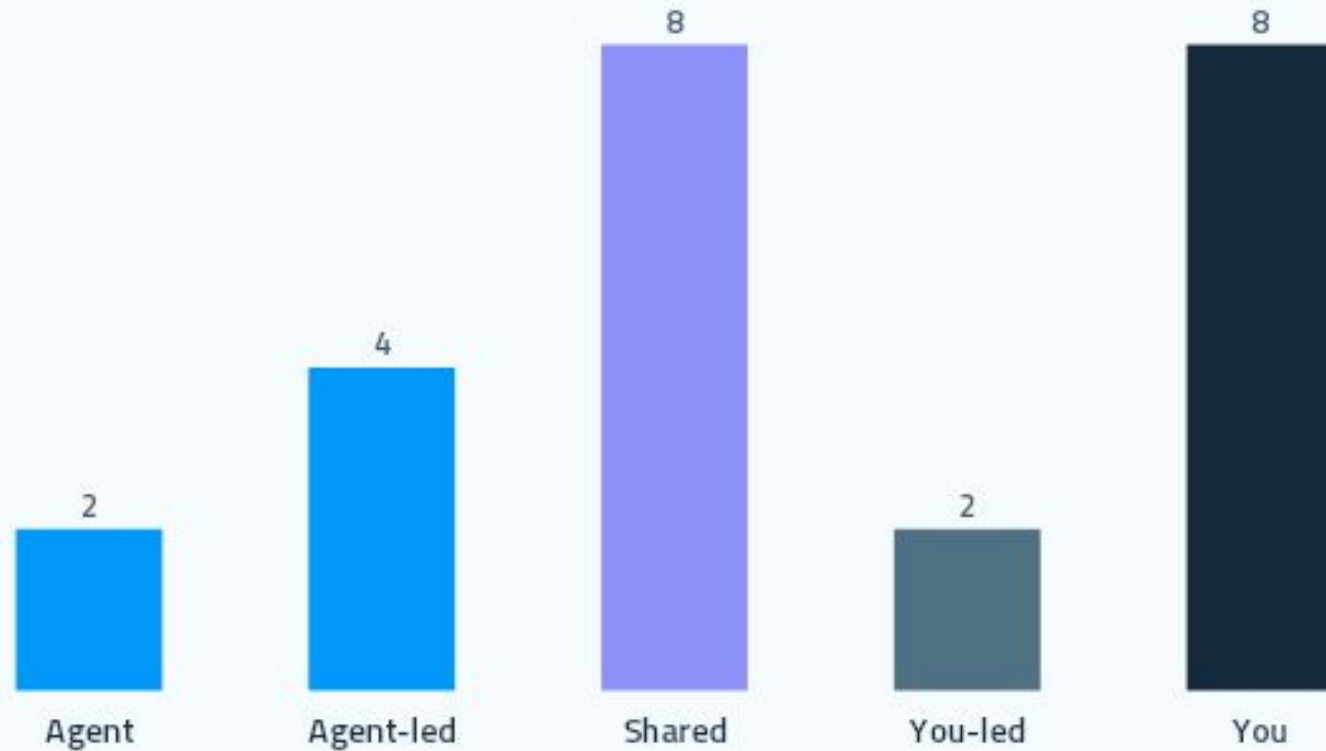
STEP	WITH AI TODAY	LEVEL
Test & doc evidence	Generates the artifacts	Agent-led
Risk assessment	Runs riskmetric/risk.assessr, assembles evidence	Shared
Validation pack	Assembles and formats	You
Decision traceability	Records why nowhere by default	You

WHAT STAYS HUMAN

Someone signs it

The evidence can be assembled by a machine. The accountability cannot be. We come back to this properly later.

So how much of the job is actually delegable?



24 steps.

Two of them you can simply hand over. Everything with consequences lands in the middle, which means a human is in the loop by definition, not by preference.

And the workflow gained steps

NEW

CLAUDE.md

Your conventions, written down once so you stop repeating them.

NEW

Skills

Repeated workflows codified so they run the "*same way*" every time.

NEW

Decision provenance

Recording why, because the agent records it nowhere.

AI removed typing. It added curation.

Let's put it into action

Building a simple R package from scratch

Let's build a package that exports two plots

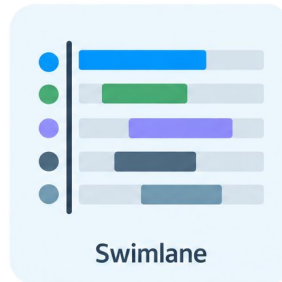
Plots that a medical monitor looks at first

Swimlane

One row per subject

- dosing periods and dose changes
- adverse events on the same timeline
- study day on the x axis
- one lane per USUBJID

answers: what happened, and when



Waterfall

One bar per subject

- best percentage change from baseline
- sorted worst to best
- coloured by response category
- reference lines at +20 and -30

answers: who responded, and by how much



Neither plot is hard. The package is not about the maths.

It starts with one plot

A medical monitor, mid-study, with `adtr_onco` and no chart

01

THE MEDICAL MONITOR

Best response, as a waterfall



WHAT WE BUILT

One function. `ggplot2`. One test.

ADDS

Nothing here is wrong. This is a good little package.

Then a second plot

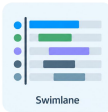
The same monitor, now wanting dosing and adverse events on one timeline

01	THE MEDICAL MONITOR Best response, as a waterfall	 Waterfall	WHAT WE BUILT One function. ggplot2. One test.	ADDS
02	THE SAME MONITOR Dosing and events, as a swimlane	 Swimlane	WHAT WE BUILT A second function, tidying copied across	ADDS

Still fine. The second plot fits the shape of the first.

Then people you have never met

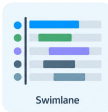
A Shiny team building a monitoring dashboard, who need interactivity

01	THE MEDICAL MONITOR Best response, as a waterfall	 Waterfall	WHAT WE BUILT One function. ggplot2. One test.	ADDS
02	THE SAME MONITOR Dosing and events, as a swimlane	 Swimlane	WHAT WE BUILT A second function, tidying copied across	ADDS
03	A SHINY TEAM The same charts, but interactive		THE EASY ANSWER engine = "plotly", a switch in both	CHANGES SHAPE

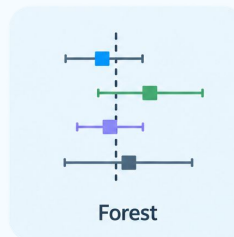
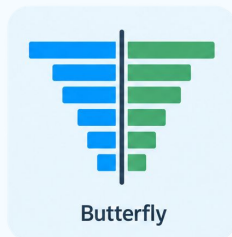
Not wrong. Just the average answer, arriving already tested.

Then the ground moves underneath

A legacy study that never mapped to ADaM, reading SDTM domains directly

01	THE MEDICAL MONITOR Best response, as a waterfall	 Waterfall	WHAT WE BUILT One function. ggplot2. One test.	ADDS
02	THE SAME MONITOR Dosing and events, as a swimlane	 Swimlane	WHAT WE BUILT A second function, tidying copied across	ADDS
03	A SHINY TEAM The same charts, but interactive		THE EASY ANSWER engine = "plotly", a switch in both	CHANGES SHAPE
04	A LEGACY STUDY The same charts, from SDTM not ADaM		THE EASY ANSWER standard = "sdtm", a second switch	CHANGES SHAPE

Two switches now, inside every plot function. Nobody decided this.



 Create the **'floaties'** package that exports swimlane and waterfall plot generators in *ggplot2*, *plotly*, and *echarts4r*

Fork and raise a PR in this repository:
<https://github.com/vedhav/floaties>